Week 1 – Assignment 2 **Mar 31 2026**

Assignment Q 1) From the Groq and Openairouter

https://console.groq.com/home

https://developers.openai.com/api/docs/models/all

--List down the Multimodels

From Groq site

List of Multi Modal Models from Groq site

Speech to Text

1) Whisper: whisper-large-v3
2) Whisper : whisper-large-v3-turbo

Text to Speech

3) Canopy Labs Orpheus V1 English: canopylabs/orpheus-v1-english
4) Canopy Labs Orpheus Arabic Saudi: canopylabs/orpheus-arabic-saudi

Vision

5) Llama 4 Scout 17B 16E: meta-llama/llama-4-scout-17b-16e-instruct

From Open AI site

https://developers.openai.com/api/docs/models/all

1) GPT 5 is multi modal
2) Image processing: These are multi modal
    a. GPT Image 1.5
    b. GPT Image 1
    c. DALLE 3
    d. DALLE 2
    e. Gpt-image-mini
    f. Chatgpt-image-latest

3) Video processing: These are multi modal

a. Sora 2
b. Sora 2 Pro

4) Audio: These are multi modal

Realtime & audio Models for realtime, speech, and audio workflows.

gpt-realtime-1.5
The best voice model for audio in, audio out

gpt-realtime
Model capable of realtime text and audio inputs and outputs

gpt-realtime-mini
A cost-efficient version of GPT Realtime

gpt-audio-1.5
The best voice model for audio in, audio out with Chat Completions.

gpt-audio
For audio inputs and outputs with Chat Completions API

gpt-audio-mini
A cost-efficient version of GPT Audio

GPT-4o Audio
GPT-4o models capable of audio inputs and outputs

GPT-4o mini Audio
Smaller model capable of audio inputs and outputs

GPT-4o Realtime
Model capable of realtime text and audio inputs and outputs

GPT-4o mini Realtime
Smaller realtime model for text and audio inputs and outputs

GPT-4o Transcribe
Speech-to-text model powered by GPT-4o

GPT-4o mini Transcribe
Speech-to-text model powered by GPT-4o mini

GPT-4o Transcribe Diarize
Transcription model that identifies who's speaking when

GPT-4o mini TTS
Text-to-speech model powered by GPT-4o mini

TTS-1
Text-to-speech model optimized for speed

TTS-1 HD
Text-to-speech model optimized for quality

Whisper
General-purpose speech recognition model

5) Coding – latest ones are multi modal:
a. GPT-5-Codex
b. GPT-5.1-Codex
c. GPT-5.2-Codex
d. GPT-5.3-Codex
e. GPT-5.1-Codex Max
f. GPT-5.1-Codex Mini
g. Codex-mini-latest

Assignment Q 2) Take 3 models from each service provider

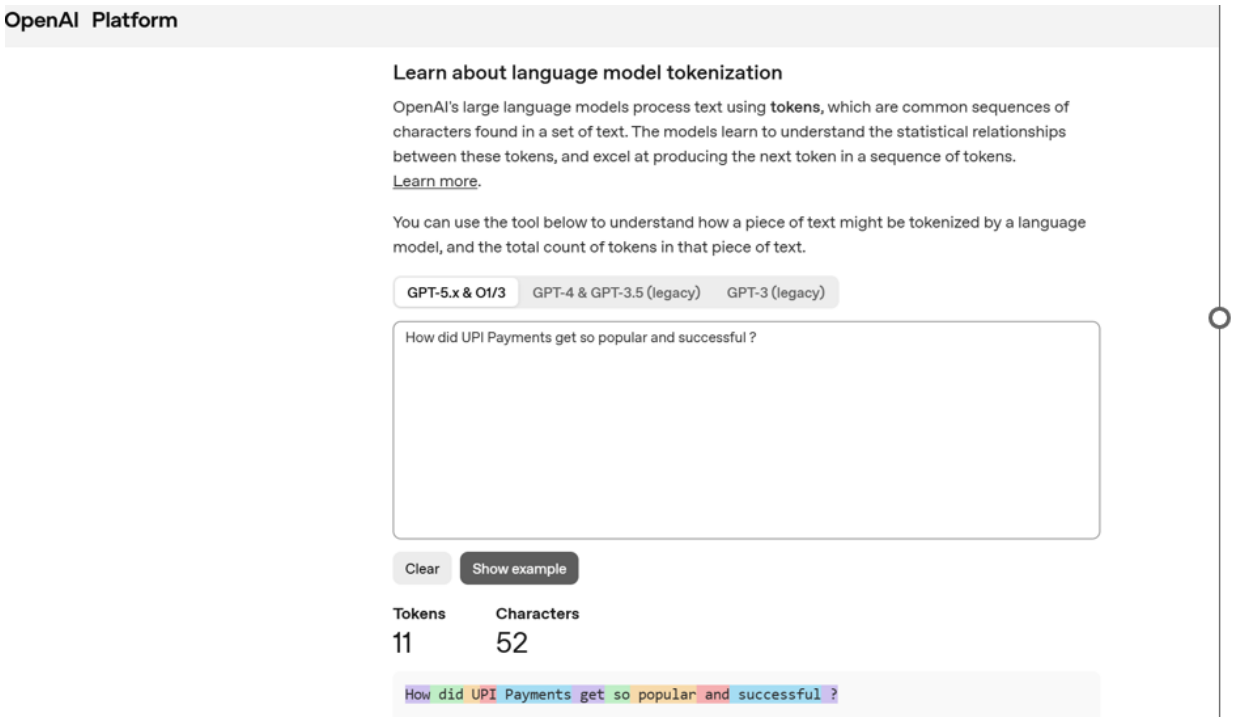
https://openrouter.ai/models

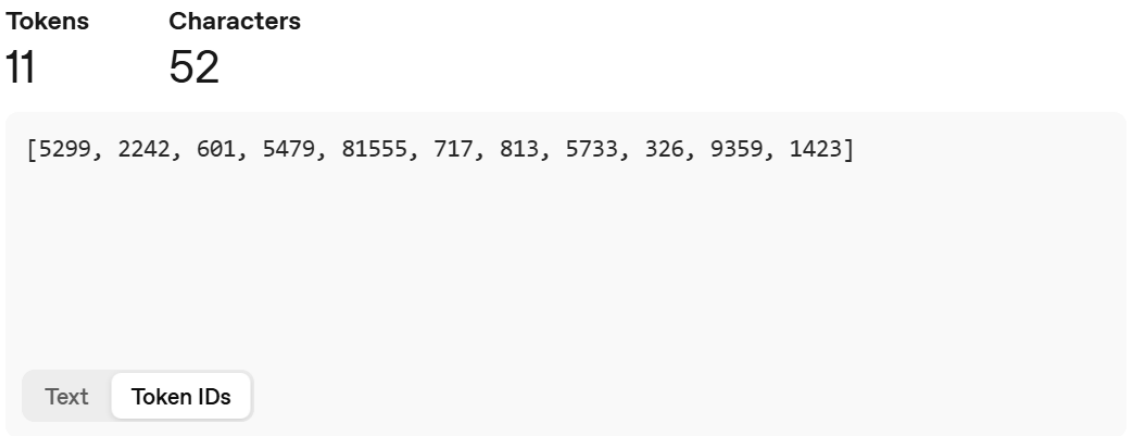
Context window,Input output token cost

Service Provider	Model	Context Window	Input Token Cost	Output Token Cost	Audio Tokens
Open AI	GPT- OSS - 120B	131K context	$0.039/M input tokens	$0.19/M output tokens	
	GPT - 5.4	1.05M context	$2.50/M input tokens	$15/M output tokens	
	GPT- 5.3 - Codex	400K context	$1.75/M input tokens	$14/M output tokens	
Google	Gemini 2.5 Pro	1.05M context	$1.25/M input tokens	$10/M output tokens	$1.25/M audio tokens
	Gemini 2.5 Flash Lite	1.05M context	$0.10/M input tokens	$0.40/M output tokens	$0.30/M audio tokens
	Gemini 2.5 Flash	1.05M context	$0.30/M input tokens	$2.50/M output tokens	$1/M audio tokens
DeepSeek	DeepSeek V 3.2	164K context	$0.26/M input tokens	$0.38/M output tokens	
	deepseek/deepseek-r1-0528	164K context	$0.45/M input tokens	$2.15/M output tokens	
	deepseek/deepseek-r1-distill-qwen-32b	33K context	$0.29/M input tokens	$0.29/M output tokens	
Alibaba	alibaba/tongyi-deepresearch-30b-a3b	131K context	$0.09/M input tokens	$0.45/M output tokens	
Anthropic	anthropic/claude-sonnet-4.6	1M context	$3/M input tokens	$15/M output tokens	
	anthropic/claude-opus-4.6	1M context	$5/M input tokens	$25/M output tokens	
	anthropic/claude-haiku-4.5	200K context	$1/M input tokens	$5/M output tokens	
Microsoft	microsoft/phi-4	16K context	$0.065/M input tokens	$0.14/M output tokens	
	microsoft/wizardlm-2-8x22b	66K context	$0.62/M input tokens	$0.62/M output tokens	
	microsoft/phi-4-reasoning-plus	33K context			
Meta	meta-llama/llama-3.1-8b-instruct	16K context	$0.02/M input tokens	$0.05/M output tokens	
	meta-llama/llama-4-maverick	131K context	$0.10/M input tokens	$0.32/M output tokens	
	meta-llama/llama-4-scout	328K context	$0.08/M input tokens	$0.30/M output tokens	

Assignment Q 3) use the Tokenizer with different prompt and see how the tokens are generated- https://platform.openai.com/tokenizer

Example Prompt 1





Example Prompt 2

GPT-5.x & O1/3 | GPT-4 & GPT-3.5 (legacy) | GPT-3 (legacy)

How long does it take to prepare for ISTQB?

Clear | Show example

Tokens 11

Characters 43

How long does it take to prepare for ISTQB?

Text | Token IDs

GPT-5.x & O1/3 | GPT-4 & GPT-3.5 (legacy) | GPT-3 (legacy)

How long does it take to prepare for ISTQB?

Clear | Show example

Tokens 11

Characters 43

[5299, 1701, 2226, 480, 2304, 316, 13708, 395, 39688, 156490, 30]

Text | Token IDs

Example Prompt 3

GPT-5.x & O1/3 | GPT-4 & GPT-3.5 (legacy) | GPT-3 (legacy)

Show details on Oscar awards last three years

Clear | Show example

Tokens 8

Characters 45

Show details on Oscar awards last three years

Text | Token IDs

GPT-5.x & O1/3 | GPT-4 & GPT-3.5 (legacy) | GPT-3 (legacy)

Show details on Oscar awards last three years

Clear | Show example

Tokens 8

Characters 45

[8313, 4878, 402, 41929, 27153, 2174, 3407, 2101]

Text | Token IDs